HDLC — sinxron kanal protokoli

High-level Data Link Control

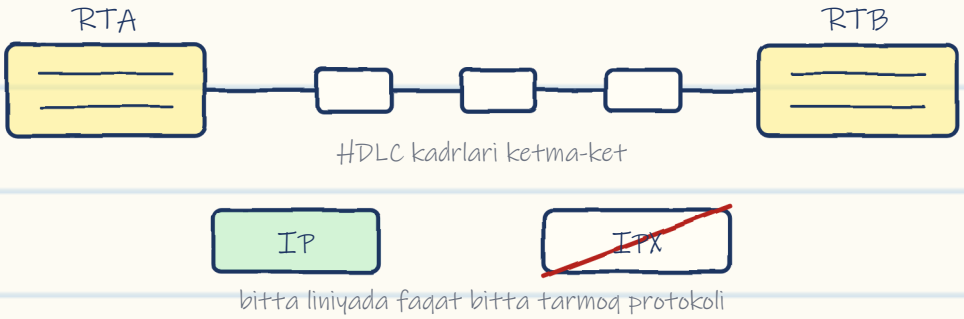
Asosiy atamalar
- HDLC — bitga yo'naltirilgan sinxron kanal protokoli (ISO)
- Flag — kadr chegarasini belgilovchi bayt 01111110 = 7E
- Bit stuffing — 5 ta '1' dan keyin '0' qo'shib yuboriladi
- FCS — kadrni tekshiruvchi 16 yoki 32 bitli CRC
- Nuqta-nuqta — ikki qurilma orasidagi sinxron seriya liniya

1. HDLC nima?

Bitga yo'naltirilgan sinxron kanal qatlami protokoli (ISO)

- asosi — IBM SDLC; ANSI ning ADCCP protokoli bilan bir xil
- shaffof — har qanday bit oqimini uzatadi, kod jadvaliga bog'liq emas
- faqat sinxron nuqta-nuqta liniyalar: serial sinxron rejim, POS

sinxron seriya liniyasi · nuqta-nuqta · full duplex



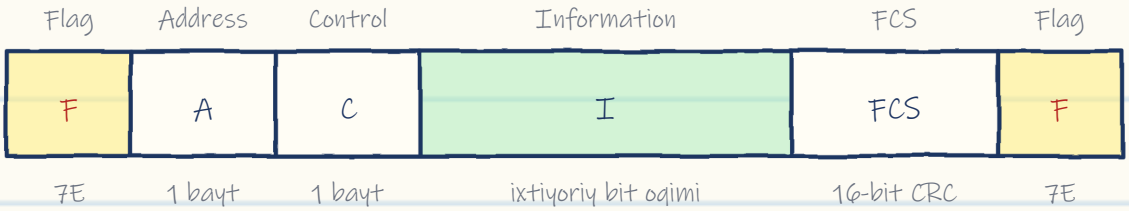
1-rasm. Yuqori protokolni belgilovchi maydon yo'q

2. Kadr chegarasi va bit stuffing



2-rasm. Flag = 01111110 = 7E

3. Kadr formati

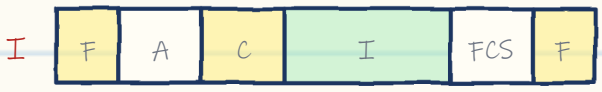


har kadr F bilan boshlanadi va F bilan tugaydi

3-rasm. Data ham, boshqaruv ham — shu format

- F — kadr boshi va oxiri
- A — kadr manbasi yoki maqsadi
- C — kadr turi: boshqaruv yoki data
- I — data, ixtiyoriy ikkilik ketma-ketlik
- FCS — 16 bitli CRC, kadr tarkibini tekshiradi

4. Kadr turlari



data + oqim va xato nazorati



xato va oqim nazorati

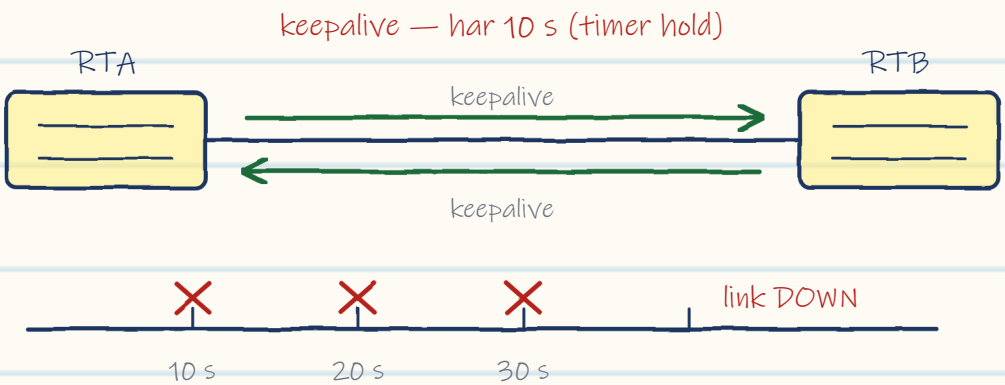


kanalni o'rnatish va uzish

S va U kadrlarda I maydoni yo'q

Eslatma: I-kadrlar raqamlanadi — yo'qolgan yoki takroriy kadr aniqlanadi.

5. Kanal holatini nazorat

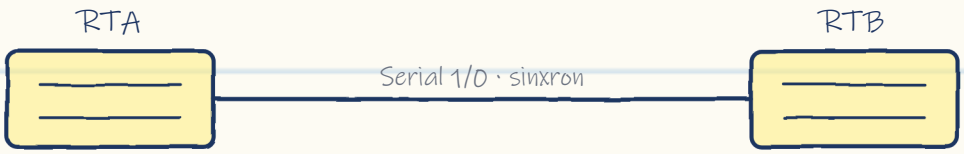


javob kelmagan intervallar ketma-ket sanaladi

4-rasm. Fizik qatlam tayyor bo'lgach keepalive boshlanadi

- timer hold — standart 10 s, oraliq 0–32767 s
- ikki tomonda bir xil qiymat bo'lsin
- ikki tomonda 0 — holat nazorati o'chadi

6. Sozlash



ikki tomonda bir xil

```
[RTA-Serial1/0] link-protocol hdlc
[RTA-Serial1/0] timer hold 10
```

```
[RTB-Serial1/0] link-protocol hdlc
[RTB-Serial1/0] timer hold 10
```

timer hold 0 — nazorat o'chadi

link-protocol hdlc	sinxron interfeysda HDLC
timer hold sekund	keepalive intervali, 0–32767

Eslatma: serial interfeysning standart kanal protokoli — PPP.

7. Kuchli va kuchsiz tomonlari

kuchli tomonlari

- ✓ shaffof bit uzatish
- ✓ full duplex — samarali
- ✓ FCS hamma kadrda bor
- ✓ I-kadrlar raqamlanadi
- ✓ yagona kadr formati

cheklovlari

- ✗ autentifikatsiya yo'q
- ✗ IP manzil kelishuvi yo'q
- ✗ faqat sinxron nuqta-nuqta
- ✗ ko'p protokolni tanimaydi

Xulosa

- HDLC — ISO, bitga yo'naltirilgan, sinxron
- Kadr: F A C I FCS F, chegara 7E
- 5 ta '1' dan keyin '0' — bit stuffing
- I · S · U — data, nazorat, kanal boshqaruvi
- keepalive: timer hold, standart 10 s
- Xavfsizlik va IP kelishuvi yo'q

Tekshirish

1. Flag qiymati? — 01111110 = 7E
2. FCS uzunligi? — 16 bit CRC
3. Standart polling interval? — 10 s
4. HDLC ni yoqish? — link-protocol hdlc
5. Kadr turlari? — I, S, U